



INTRODUCTION TO BIOSTATISTICS

Saad Ibrahim Al-Ghabban

Learning objectives

By the end of this lecture the student will be able to:

- Define key statistical terms such as *data*, *variable*, *population*, *sample*, *descriptive statistics*, and *inferential statistics*.
- List the types of data (quantitative and qualitative) and variables (numerical and categorical).
- Describe the differences between descriptive and inferential statistics.
- Summarize the characteristics of nominal, ordinal, discrete, and continuous variables.
- Classify given examples of data into appropriate types (e.g., discrete vs. continuous, nominal vs. ordinal).
- Distinguish between types of variables

Statistics vs Biostatistics

Statistics is the science that deals with

1. The methods of collection, organization, analysis, interpretation, and presentation of information that can be stated numerically (data).
2. The drawing of inferences about a body of data when only a part of the data is observed.

Biostatistics is a subset of statistics used to handle health-relevant information

Objectives

1. to learn how to organize and summarize data
2. to learn how to reach decisions about a large body of data by examining only a small part of the data

Data

Data is a collection of facts, observations, measurements, or information that are recorded for the purpose of analysis, to answer questions or make decisions. In simple terms, data are the raw pieces of information before they are organized and interpreted.

Data may be

- a. **Qualitative data**
 - Data in the form of words
 - No tool used to measure it
 - Does not describe quantities or amounts
 - e.g.: age, weight, height, family income, cell count, body temperature, number of births, blood pressure , etc
- b. **Quantitative data**
 - Describes quantities expressed numerically
 - There is an objective way of measuring or counting it
 - e.g.: gender, race, residence education, socioeconomic status, marital status, levels of satisfaction etc.

Quantitative Data	Qualitative Data
• Deals with numbers. • Data that be measured, counted, or recorded.	• Deals with descriptions. • Observed but not measured.

Data collection

1. Primary data
 - Data collected from the respondent, for the first time by the investigator for a specific purpose
 - No statistical operations have been performed
 - Pure data/ source data
2. Secondary data
 - Data sourced from somewhere e.g. hospital records, other's research results
 - Statistical analyses have already been performed on it

Variables

A variable (also known data items) is an attributes of an object under study. It can be any characteristic (aspect) of the individual that varies among the different members of the population. ⁽¹⁾. It can be **measured** (like blood pressure), **counted** (like number of patients), or **recorded**, (like age or gender).

It takes on different values in different persons, places, or times; the reason is that the characteristic is not the same when observed in different possessors of it. Blood pressure varies among different people; it also varies among different times for the same person

Types of Variables⁽²⁾

Variables have types depending on the data they contain

1. **Numerical (Quantitative) variables** include counts, (e.g. the number of children) and measurements (e.g. weight of children). They are either discrete or continuous:
 - a. Discrete variables: counts of individual items, characterized by only a **countable number** of distinct values, they represent data with an assigned order and magnitude but are restricted to whole numbers or integers; examples include (e.g. no. of children, number of pregnancies, number of attacks of asthma per week). Qualitative data are discrete.
 - b. Continuous variables: measurements of individual items, they are those, which are **measured** on a continuous **scale**, they represent data capable of possessing any value in a given range, such as blood pressure, temperature, age, weight, Hb leveletc.). Quantitative data are discrete or continuous.
2. **Categorical (Qualitative) variables** are data that one can name and put into categories. They are not capable of being measured. Each individual is one of a number of mutually exclusive classes. They are discrete, always.
 - a. Nominal data: categories are unordered. Variables are not related to each other & cannot be arranged.
 - 1) *Dichotomous (binary)*: Two mutually exclusive categories e.g., gender, survival, exposure state.
 - 2) *Multichotomous*: blood group, marital status, occupation eye colour, etc.
 - b. Ordinal data: Categories are ordered, they express ranks and have natural order or progression
 - 1) *Score ranking*: birth order.
 - 2) *Category ranking* (ranking for group of responses) e.g., disease stage (mild-moderate- severe), social class (high, middle, low), education level (primary school, secondary school, university)

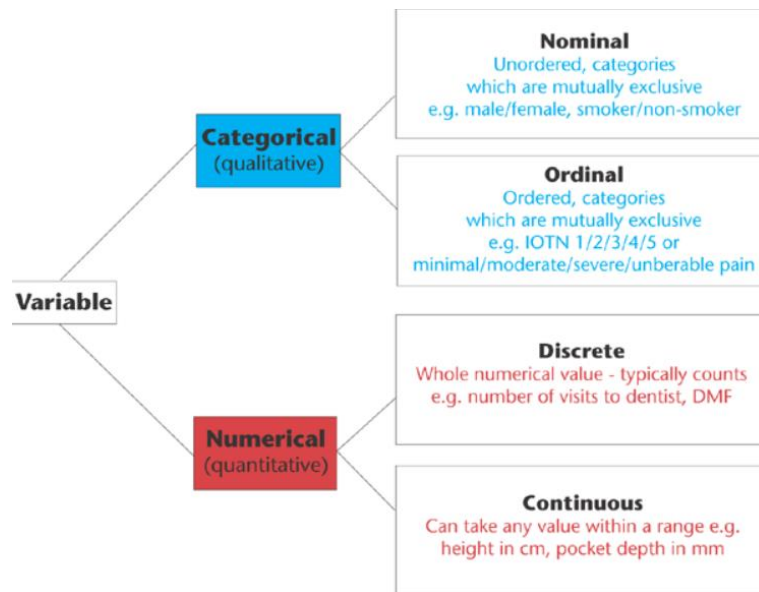


Figure 1: Types of variables

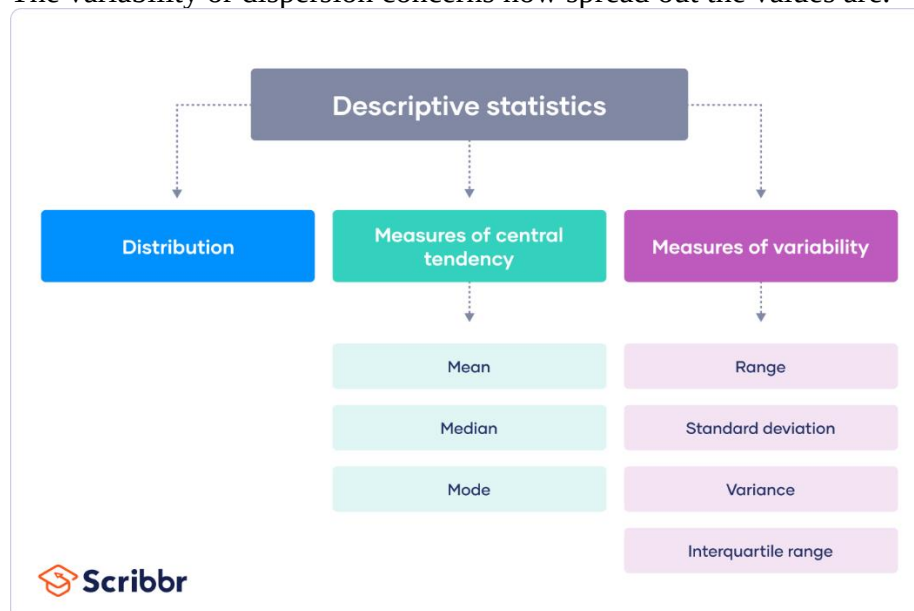
Types of statistics:

1. **Descriptive Statistics:** that summarize or describe the important characteristics of a known set of population data. They are methods of producing quantitative summaries of information (measures of central tendency, measures of dispersion).

Descriptive statistics summarize and organize characteristics of a data set. A data set is a collection of responses or observations from a sample or entire population.

Using descriptive statistics, you can report characteristics of your data:

- a. The distribution concerns the frequency of each value.
- b. The central tendency concerns the averages of the values.
- c. The variability or dispersion concerns how spread out the values are.



Researchers use descriptive statistics of data collected on a “Sample” from a much larger group called the “Population”.

2. **Inferential Statistics:** that use sample data to make inferences (or generalizations) about a population. Inferential statistics are guesses (estimates or decisions) about a population based on sample data from that group, with an evaluation of the certainty of the guess.

Statistic inference is the process of drawing conclusions regarding a population based upon studying only a portion of the members of that population. This process of inference (i.e., generalization), is most accurate when the sample (subpopulation) being studied is :

- a. homogenous (i.e., uniform in character).
- b. truly representative (i.e., correctly reflects the larger group), and
- c. chosen randomly to eliminate sampling error or bias.

While descriptive statistics summarize the characteristics of a data set, inferential statistics help you come to conclusions and make predictions based on your data. When you have collected data from a sample, you can use inferential statistics to understand the larger population from which the sample is taken.

Inferential statistics have two main uses:

- a. Making estimates about populations (for example, the mean BMI in adults).
- b. Testing hypotheses to draw conclusions about populations (for example, the relationship between BMI and family income).

In quantitative research, after collecting data, the first step of statistical analysis is to describe characteristics of the responses, such as the average of one variable (e.g., age), or the relation between two variables (e.g., age and creativity).

The next step is inferential statistics, which help you decide whether your data confirms or refutes your hypothesis and whether it is generalizable to a larger population.

Population and Sample

The population (also called reference, target, or study population) is the entire wide group that you want to study and draw conclusions and make an inference about ⁽³⁾. It is not limited to a population of people but can refer to any collection of objects (an aggregate of creatures, things, cases and so on). For example, the data may relate to a sample of 20 hospitals from the population of all hospitals in the country.

When you conduct research about a group of people, it is rarely possible to collect data from every person in that group. Instead, you select a sample. Care must be taken to ensure that the sample truly represents the “target population” about which information is required.

The sample is the specific group of individuals who will actually participate in the research and you will use descriptive statistics to collect data from ⁽³⁾.

It may be defined as a subgroup drawn from a population (it consists of selected members of a population). The sample is of interest not in its own right, but for what it tells the investigator about the population.

Statistics allows us to use the sample to make inferences about the population from which it was derived. Because of chance, different samples from the population will give different results.

Examples:

From 5000 Undergraduate students in the university (as for Population), 300 undergraduate students from three colleges who volunteer for your research study (as for the sample), were selected

References

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